

Household & bicycle maintenance

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1. Household & bicycle maintenance

Practical checklists and maintenance notes for setting up a home, gardening on a balcony or small plot, and keeping a road bike in good working order.

The book is organised in three parts:

1. **Gardening and planting** — containers, substrate, crops, and basic tools.
2. **Interior DIY and equipment** — room-by-room shopping and setup lists for moving in.
3. **Bicycle maintenance** — anatomy, brakes, gears, wheels and tyres, and steering adjustment.

All lists use GitHub-style task checkboxes (- [] / - [x]) so you can tick items off as you go.

Figure 1.1.: High-level structure of this book.

Part I.

Gardening and planting

2. Gardening checklists

Equipment and planting lists for balcony or small-space gardening. Incorporated from issue #2.

2.1. Containers

- ☐ Flower pots (assorted sizes)
- ☐ Window boxes for balcony ledges
- ☐ Deep grow bags / planter tubs (ideal for tomatoes), **30 L** minimum
- ☐ Hanging pots to save space
- ☐ Saucers under pots (to catch runoff)

2.2. Substrate

- ☐ Potting compost
- ☐ Compost (soil improver)
- ☐ Mulch (straw or bark chips)
- ☐ Liquid vegetable fertiliser

2.3. Vegetables and herbs

- ☐ Cherry tomatoes, radishes, lettuces, chillies or small peppers
- ☐ Aromatic herbs: basil, parsley, thyme, rosemary, mint, chives

2. Gardening checklists

- ☐ Useful or decorative flowers: nasturtiums, French marigolds, petunias, geraniums

2.4. Supports

- ☐ Stakes or netting for climbing plants

2.5. Gardening tools

- ☐ Trowel (hand transplanting spade)
- ☐ Hand cultivator (garden claw)
- ☐ Watering can
- ☐ Spray bottle (for misting leaves)

Figure 2.1.: Simple planting workflow for balcony crops.

Part II.

Interior DIY and equipment

3. Interior equipment lists

Room-by-room checklists for furnishing and stocking a flat. Incorporated from issue #1.

3.1. Temporary actions

- ☐ Wardrobe kit with curtain

3.2. Kitchen furniture and appliances

- ☐ Fridge + freezer
- ☐ Microwave
- ☐ Oven
- ☐ Hob (induction adapter if needed)
- ☐ Kitchen table + chairs
- ☐ **Measuring jug**
- ☐ Coffee maker
- ☐ Kettle
- ☐ Placemats, boards
- ☐ Bowls, glasses, mugs
- ☐ Cutlery, and spatulas (wood and plastic) / ladle
- ☐ **Plates**
- ☐ Mixing bowl (salad bowl)
- ☐ **Frying pans**

3. *Interior equipment lists*

- ☐ Saucepans
- ☐ **Lids**
- ☐ Vegetable peeler
- ☐ Electric hand mixer
- ☐ Blender
- ☐ Food processor
- ☐ Grater
- ☐ **Colander**
- ☐ Draining rack
- ☐ Tea towels
- ☐ Washing-up liquid
- ☐ **Soap**
- ☐ Kitchen scales
- ☐ **Whisk**
- ☐ Aluminium foil, baking parchment, baking paper
- ☐ **Knife sharpener**
- ☐ Bin + bin bags
- ☐ **Chopping board**

3.3. **Food (to get started)**

- ☐ Oil, spices (pepper, curry, paprika), salt
- ☐ Garlic, onions, shallots
- ☐ Potatoes
- ☐ Ice packs
- ☐ Flour, sugar
- ☐ Pasta, rice
- ☐ Pâté, cured sausages
- ☐ Jam
- ☐ Baking kit (vanilla, yeast)
- ☐ **Honey**
- ☐ **Coffee, herbal teas**

3.4. Bedding and bedroom

- ☐ **Single-portion tinned goods**

3.4. Bedding and bedroom

- ☐ Wardrobe for clothes + hangers
- ☐ Bed (base + mattress)
- ☐ Duvet and sheet
- ☐ Pillows and pillowcases
- ☐ Bedside table
- ☐ Lamp (**second lamp**)
- ☐ Camp beds
- ☐ Sofa bed (fold-out)
- ☐ Shoe storage furniture
- ☐ **Plastic crates / stacking towers**

3.5. Bathroom and shower

- ☐ Brush
- ☐ Mop (bucket) + dustpan and brush
- ☐ Vacuum cleaner
- ☐ Bath mat
- ☐ First-aid kit + sun cream
- ☐ Plasters + antiseptic
- ☐ Toothbrush, nail clippers, safety pin, toothpaste
- ☐ Razor (manual + electric), shaving foam
- ☐ Bathroom scales
- ☐ Floor cleaner
- ☐ Toilet cleaner (drains) and/or limescale remover
- ☐ (wood cleaner)
- ☐ Glass cleaner
- ☐ White vinegar + bicarbonate of soda

3. Interior equipment lists

- ☐ Laundry detergent
- ☐ Washing machine
- ☐ Clothes airers
- ☐ Shower curtain, rail
- ☐ Laundry basket
- ☐ Toilet paper
- ☐ Gloves and towels

3.6. Clothing and sport

- ☐ Underwear
- ☐ Rain clothing
- ☐ Sportswear
- ☐ Summer clothing
- ☐ Casual wear (baggy trousers, jumpers)
- ☐ Suit
- ☐ Shoes (*3 pairs*)
- ☐ Hydration pack
- ☐ Bicycles + locks + helmet + gloves + cap
- ☐ Water bottles and thermos flasks
- ☐ Slippers and flip-flops
- ☐ Swimming gear

3.7. Work

- ☐ Computers
- ☐ Extension lead and chargers
- ☐ Graphics tablet
- ☐ Notebooks, pencil case, **stamps**, envelopes
- ☐ Desk, lamp, office chair
- ☐ External hard drive, power bank

- ☐ Headphones

3.8. Leisure

- ☐ Electronic kit (chargers, Ethernet cable, ...)
- ☐ Hiking rucksack
- ☐ Hiking equipment (head torch, ...)
- ☐ Bicycle, bottle, repair kit, track pump
- ☐ DIY tools
- ☐ Books + bookcase
- ☐ **Board games**, playing cards

Part III.

Bicycle maintenance

4. Bike overview and equipment

General anatomy notes, pedal fitting, and the toolkit required for routine maintenance. Based on personal notes (`vélo astuces.docx`).

4.1. Pedal orientation and thread direction

The left pedal sits on the opposite side of the bottom bracket shell (i.e. on your left when you mount the bike).



Figure 4.1.: Which way to turn a screw

- **Fitting:** the right pedal tightens clockwise; the left pedal tightens anti-clockwise.
- **Removal:** the right pedal loosens anti-clockwise (the opposite of normal right-hand threads).

4. Bike overview and equipment

Figure 4.2.: Pedal thread directions (viewed from the outside of the crank).

4.2. Essential toolkit

For a Shimano 105 triple 10-speed road bike (5700/5703 generation) with an FSA Gossamer crankset, the following tools cover most jobs.

Tool	Typical use	Status
Chain tool (6–11 s)	Shorten or remove chain	In Decathlon 90
Hex keys	Derailleurs, chainrings	In box
Chainring nut wrench	Hold nuts behind chainrings	In box
Torx T30	Some chainring bolts	In box
Chain whip + Shimano HG lockring tool	Cassette removal	In box
10 mm hex key / bit	FSA 30 mm crank bolt	Add
Torque wrench (41 Nm)	Crank bolt at 38–41 Nm	Add
Quick-link pliers	Chains with a master link	Recommended
Bicycle grease	Axle, threads, chainring bolts	Recommended
Cable cutters	Only if a cable is frayed	Optional

Decathlon’s published 900-box contents do not list a 10 mm hex key; their separate hex set stops at 8 mm — **check your physical box before buying.**

4.2.1. Torque wrench note

For an FSA 30 mm axle with a self-extracting bolt, FSA specifies a **10 mm hex** and **38–41 Nm** torque with a calibrated wrench.

A compact 4–24 Nm wrench with a 10 mm bit is excellent for derailleurs and chainring bolts but **does not reach crank-bolt torque.** Prefer a

4.3. Bike anatomy (road)

3/8 or 1/4 + 3/8 wrench covering roughly **5–50 Nm or 5–60 Nm** with a **10 mm hex bit**.

4.3. Bike anatomy (road)

Figure 4.3.: Main components referenced in this part of the book.

4.4. Required riding equipment

- ☐ Helmet
- ☐ Front and rear lights
- ☐ Lock(s)
- ☐ Gloves
- ☐ Water bottles or hydration pack
- ☐ Mini pump or CO₂ inflator
- ☐ Spare inner tube and tyre levers
- ☐ Multi-tool and quick-link
- ☐ Mudguards (wet weather)

4.5. Crankset removal (after pedals)

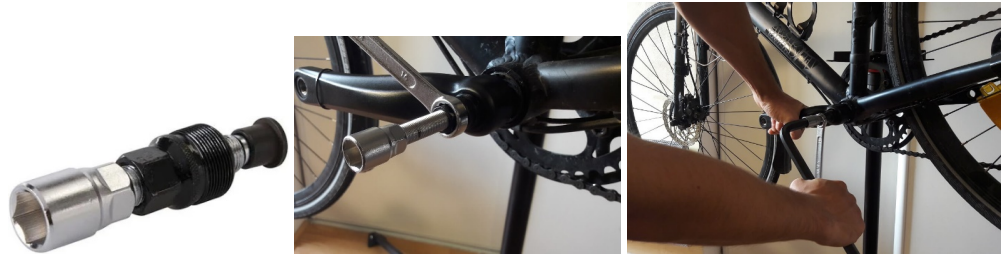
Traditional square-taper or external-bearing systems use a crank extractor; **FSA 30 mm self-extracting axles use a 10 mm hex in the central bolt instead** — see Section 6.4 for the full chainring-replacement workflow.

For extractor-type cranks:

1. Remove the chain.

4. *Bike overview and equipment*

2. Remove the retaining bolt at the base of the crank.
3. Fit the crank extractor; if stiff, back the extractor off slightly and tap with a mallet.
4. Tighten the extractor with a 14 mm spanner until the crank releases.
Repeat on both sides, turning anti-clockwise.



Reference videos:

- Road crank removal

Figure 4.4.: Crank removal with a classic extractor (non self-extracting axles).

5. Brakes

Brake types, inspection, and lever replacement notes.

5.1. Brake caliper types

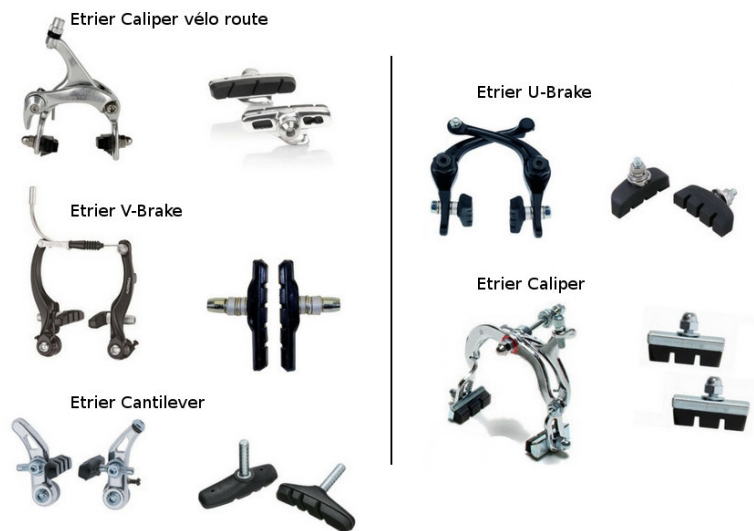


Figure 5.1.: Overview of common bicycle brake calipers

Common road-bike families:

5. Brakes

Type	Where	Monitoring focus
Rim (dual-pivot, side-pull)	Frame/fork bridges	Pad wear, rim surface, cable friction
Cantilever / V-brake	Older hybrids, CX	Pad alignment, straddle cable
Disc (mechanical or hydraulic)	Modern road, gravel	Rotor thickness, pad gap, fluid level

5.2. How to monitor rim brakes

1. **Pad thickness** — replace before the metal holder contacts the rim.
2. **Pad alignment** — toe-in slightly at the leading edge; full contact without rubbing the tyre.
3. **Rim wear** — look for a wear groove disappearing or a concave profile.
4. **Cable and housing** — smooth lever travel; no fraying at the end.
5. **Quick test** — wheel spins freely off the stand; firm lever feel with ~2–3 cm travel before firm contact.

Figure 5.2.: Rim-brake inspection routine.

5.3. Brake lever replacement

Useful references when replacing or refurbishing STI levers:

- Lever replacement (video)
- Full cable and lever change
- Shimano RSX lever rebuild
- Shimano 105 ST-5600 diagram

5.3. Brake lever replacement



Figure 5.3.: STI lever internals diagram

Figure 5.4.: STI lever replacement outline.

6. Gears

Derailleur adjustment, chain cleaning, and a chainring / rear-derailleur replacement workflow for a Shimano 105 triple 10-speed bike.

Tighten = clockwise · **Loosen = anti-clockwise** (see Figure 4.1 in the overview chapter).

6.1. Front derailleur

Reference: Citycycle front derailleur guide.

1. **Height and alignment** — outer cage plate ~2 mm above the big chainring teeth; cage parallel to the chainrings.
2. **Limit screws** — with the cable slack:
 - **L (low / inner)** — loosen to move toward the frame (e.g. if it sits on the middle ring instead of the inner); tighten to move outward if the chain drops off the inner ring.
 - **H (high / outer)** — sets the outer limit on the big ring.
3. **Cable tension** — leave ~2 turns of barrel-adjuster range; if shifts to larger rings are sluggish, increase cable tension; if the chain overshoots, reduce tension.

Figure 6.1.: Front derailleur adjustment order.

6. Gears

6.2. Rear derailleur

Reference: VCMB rear derailleur guide and video walkthrough.

1. Confirm the derailleur hanger is straight and the cage aligns with the sprockets.
2. **H limit (high gear, smallest sprocket)** — turning **out** (anti-clockwise) moves the upper jockey wheel outward; turning **in** moves it toward the frame.
3. **L limit (low gear, largest sprocket)** — the opposite sense to H: turning **in** moves outward; turning **out** moves toward the frame. (The L screw is sometimes called the “high stop” in shop jargon — confusing, but common.)
4. **Cable tension** — middle chainring, smallest sprocket; increase tension if upshifts are hesitant, decrease if the chain climbs too eagerly.
5. **B-tension screw** — sets gap between upper jockey wheel and largest sprocket.

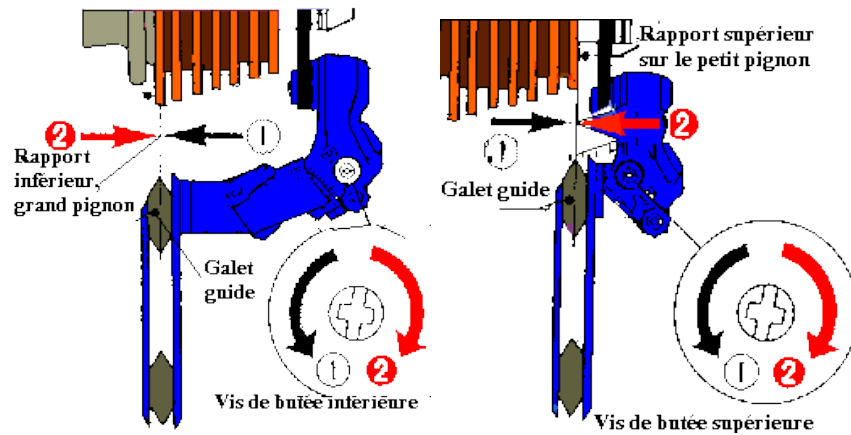


Figure 6.2.: Rear derailleur adjustment order.

6.3. Chain cleaning

1. Degrease (drivetrain cleaner or solvent).
2. Wash with warm water and washing-up liquid; remove the chain if heavily contaminated.
3. Dry thoroughly, then lubricate sparingly on each link; wipe excess.

Sources:

- YouTube chain clean
- Decathlon chain care

Figure 6.3.: Chain cleaning workflow.

6.4. Chainring and rear-derailleur replacement

Notes for an **FSA Gossamer triple 52/39/30** (five-arm: 130 mm BCD outer/middle, 74 mm BCD inner), **175 mm cranks**, **9/10-speed chain**, with **Shimano 105 ST-5703 / FD-5703** shifters.

Compatible rear derailleur: **Shimano 105 RD-5701-GS** (preferred) or **Tiagra RD-4601-GS**. Do **not** fit a Deore M6000 10-speed derailleur — cable pull differs from 5700-generation road levers.

Before ordering: confirm the largest cassette sprocket (officially 25–30 teeth for RD-5701-GS). Examples 11-28, 12-28, 12-30 are fine; 11-32 needs checking.

6.4.1. Suggested chainring parts (verify Marketplace shipping)

6. Gears

Ring	Spec
Outer 52	TA Spécialités Alizé, 130 BCD, 5-arm, 9/10 s
Middle 39	TA Spécialités Alizé, 130 BCD, 5-arm, 9/10 s
Inner 30	TA Spécialités Zelito, 74 BCD, 5-arm, triple 9/10 s

6.4.2. Replacement procedure

Figure 6.4.: Chainring and rear-derailleur replacement (keep shifters, cables, FD).

Step-by-step:

1. **Remove the chain** — quick-link pliers if fitted; otherwise use the chain tool from the 900 box.
2. **Remove the rear derailleur** — slacken the cable pinch bolt only; no need to pull the inner wire from the shifter. Unbolt from the hanger (**8–10 Nm** on refit).
3. **Set the H limit** on the new derailleur before attaching the cable.
4. **Remove the FSA crank** — left side: **10 mm hex** in the central self-extracting bolt, turn anti-clockwise. Do not remove the outer retaining ring; it is part of the extraction mechanism.
5. **Photograph washers** — especially the wave washer; keep the BB30 bearings in the frame.
6. **Replace chainrings** on the bench — use the chainring nut tool; orient markings and anti-drop pin on the outer ring behind the crank arm.
7. **Grease bolt threads lightly** — follow torque specs supplied with the new rings.
8. **Reinstall the crank** — restore washers in the original order; torque the 30 mm axle bolt to **38–41 Nm** with a calibrated wrench.

6.4. Chainring and rear-derailleur replacement

9. **Adjust the rear derailleur** — H limit, cable tension, L limit, B-tension.
10. **Fit a new 10-speed chain** — match the old link count if the previous length was correct; respect directional markings.
11. **Test** — the front derailleur should need little change if tooth counts stay 52/39/30.

6.4.3. Cassette wear warning

A new chain on a heavily worn cassette can skip under load. If skipping persists on one or two sprockets despite correct adjustment, replace the cassette — the 900 box already includes the HG lockring tool and chain whip.

Minimum extra purchases: 10 mm hex key/bit and a torque wrench to **41 Nm**; add quick-link pliers and grease for convenience.

7. Tyres and wheels

Wheel truing, cassette removal, and tyre basics.

7.1. Wheel truing

7.1.1. Tools



Figure 7.1.: Spoke wrench icon

- Spoke wrench (correct size for your nipples)
- Truing stand or frame as a guide
- Dish gauge (optional) — checks rim centreing on the hub
- Lateral indicator — measures **lateral run-out** (“dish” error in the sideways plane)

7. Tyres and wheels

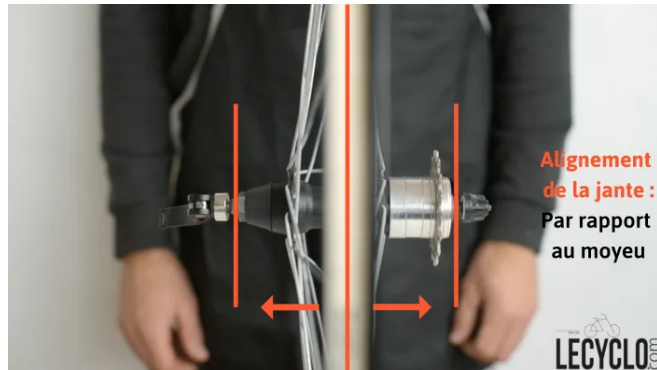


Figure 7.2.: Truing stand and indicators

7.1.2. Truing procedure

Spoke nipples tighten clockwise (viewed from the rim), increasing tension.

1. Check for broken spokes; replace any that are damaged.
2. Bring overall tension up evenly before fine truing.
3. Correct run-out in groups of **3–4 spokes** at a time:
 - If the rim pulls **left**, tighten right-side spokes and/or loosen left-side spokes.
 - Reverse for movement to the **right**.
4. Work in $\frac{1}{4}$ or $\frac{1}{2}$ **turn** increments — never force a nipple to its limit.

Full tutorial: Le Cyclo — wheel truing

Figure 7.3.: Wheel truing decision loop.

7.2. Cassette removal

Identify whether you have a **cassette** (modern freehub) or a **freewheel** (screw-on, common on 7-speed bikes).

Video: cassette removal

1. Fit the chain whip to hold the sprockets.
2. Insert the lockring tool and turn the lockring **anti-clockwise**.

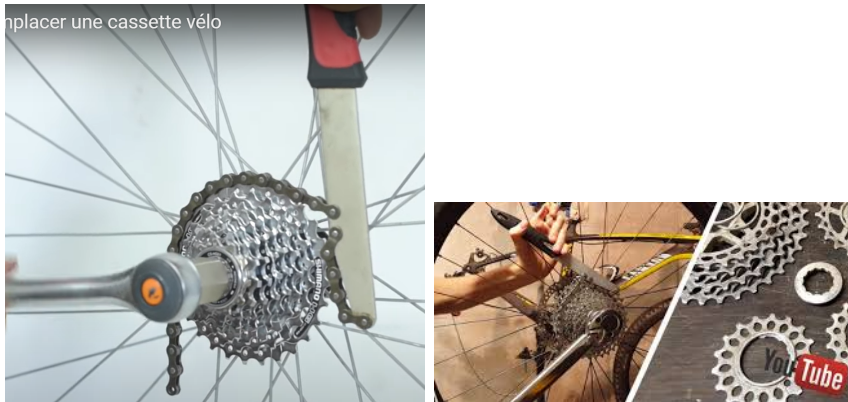


Figure 7.4.: Cassette removal on a Shimano HG freehub.

7.3. Tyres

7.3.1. Routine checks

- ☐ Correct inflation (see sidewall range; typically 6–8 bar / 80–115 psi for 25 mm road tyres)
- ☐ Tread wear indicators or flat spot in the centre
- ☐ Cuts, embedded glass, sidewall bulges

7. Tyres and wheels

- Valve seating (Presta: tighten lockring finger-tight only)

7.3.2. Tyre removal and fitting

Figure 7.5.: Road tyre replacement on the roadside or at home.

7.3.3. When to replace

Sign	Action
Frequent punctures on same wheel	Inspect rim tape and spoke ends
Squared-off tread profile	Replace tyre
Visible casing threads	Replace immediately
Age > 3–5 years (garage storage)	Consider replacement even if tread looks fine

8. Steering and headset

Headset adjustment (“steering play”) and related notes from personal maintenance logs.

8.1. Threaded headset adjustment

Excess play is felt as a knock or looseness when rocking the bike with the front brake on.

Reference video: headset adjustment

8.1.1. Adjustment sequence

1. Loosen the **locknut** (top nut).
2. Turn the **adjusting cup / bearing race** until play just disappears with the fork centred — the bars should turn smoothly without binding.
3. Hold the adjusting cup in place and re-tighten the **locknut**.
4. Re-check: no play, no notchiness.

Figure 8.1.: Threaded headset adjustment protocol.

8. Steering and headset

8.2. Integrated / threadless headsets

Most modern road bikes use a threadless system:

1. Loosen stem clamp bolts.
2. Turn the **top cap** bolt (usually 4 or 5 mm hex) to remove play — typically **0.2–0.5 Nm** preload, just enough to seat the bearings.
3. Align the stem with the front wheel.
4. Torque stem clamp bolts evenly to manufacturer spec.

Figure 8.2.: Threadless headset adjustment.

8.3. Symptoms and causes

Symptom	Likely cause
Knock when braking	Headset too loose
Notchy steering	Corroded or pitted bearings
Bars cannot centre	Bent fork or misaligned stem
Play returns quickly	Locknut not secured; wrong washer order

8.4. Related: pedal and crank orientation

Pedal thread directions and crank removal are covered in bike overview and gears. Always remove pedals before extracting cranks so the threads are not damaged by leverage on the levers.